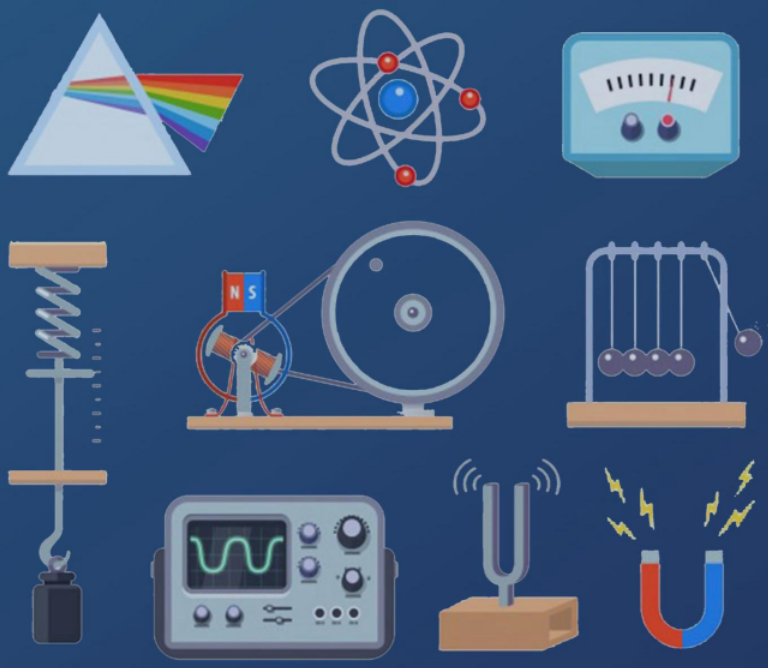
PHYSICS

Semiconductors



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-2
1	Introduction		1
2	Energy band gap	1,2,3	2
3	Intrinsic Semiconductor	4	3
4	Extrinsic Semiconductor	5,6	4,5
5	p-n junction	7,8,9,10	6-10
6	Rectifier	11,12	11,12
7	Special purpose diode	13,14	13,14
8	Transistor	15,16,17	15-19
9	Amplifier and switch	18,19,20,21	20-22
10	Logic gates	22,23,24	23-25

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Energy band gap

Q1. In semiconductor the valence band at 0 K is-

(A) completely filled

(B) completely empty

(C) partially filled

(D) nothing can be said

Q2. The energy gap in a semiconductor is of the order of :-

(A) 1 eV (B) 5 eV

(C) 10 eV (D) 15 eV

Q3. Choose the only false statement from the following :

(A) the resistivity of a semiconductor increases with increase in temperature

(B) substances with energy gap of the order of 10eV are insulators

(C) in conductors the valence and conduction bands may over lap

(D) the conductivity of a semiconductor increases with increases in temperature

Intrinsic Semiconductor

Q4. The forbidden energy gap in Ge is :-

(A) 0.72 eV

(B) 0.072 eV

(C) 7.2 eV

(D) 0.0072 eV

Extrinsic Semiconductor

Q5. P type semi conductor is :-

(A) positively charged

(B) made by mixing of impurity of boron in germanium

(C) made by mixing of impurity of phosphorus in silicon.

(D) made by mixing of impurity of carbon in silicon

Q6. In sample of pure silicon 10^{13} atom/cm^3 is mixed of phosphorus. If all doner atoms are active then what will be resistivity at 20^0C if mobility of electron is 1200 cm^2/Volt sec:-

(A) 0.5209 ohm/cm

(B) 5.209 ohm/cm

(C) 52.09 ohm/cm

(D) 520.9 ohm/cm

p-n junction

Q7. The depletion region of a p-n junction contains :–

(A) electrons only

(B) electrons and holes both

(C) holes only

(D) neither electrons nor holes

Q8. When a p-n junction is reversed biased, then the current through the junction is mainly due to :-

(A) diffusion of charge

(B) nature of the material

(C) drift of the charges

(D) both drift and diffusion of the charges

Q9. The p-n junction is a :-

(A) ohmic resistance

(B) non ohmic resistance

(C) positive resistance

(D) nagative resistance

Q10. The minority current in a p-n junction is :-

(A) From the n-side to the p-side

(B) From the p-side to the n-side

(C) From the n-side to the p-side if the junction is forward-biased and in the opposite direction if it is reverse biased

(D) From the p-side to the n-side if the junction is forward-biased and in the opposite direction if it is reverse biased

Rectifier

Q11. A device whose one end is connected to -ve terminal and the other is connected to +ve terminal, if both ends are interchanged with supply then current is not flowing then device will be :-

(A) p-n junction (B) transistor

(C) zener diode (D) triode

Q12. When two semiconductor of p and n type are brought into contact, they form a p-n junction which acts like a :-

(A) rectifier (B) amplifier

(C) oscillator (D) conductor

Special purpose diode

Q13. Zener diode is used for :-

(A) rectification

(B) stabilisation

(C) amplification

(D) producing oscillations in an oscillator

Q14. The energy of radiation emitted by LED is:-

(A) Greater than the band gap of the semiconductor used.

(B) Always less than the band gap of the semiconductor used.

(C) Always equal to the band gap of the semiconductor used.

(D) Equal to or less than the band gap of the semiconductor used.

Transistor

Q15. In transistor symbols, the arrows shows the direction of :–

(A) current in the emitter

(B) electron current in the emitter

(C) holes current in the emitter

(D) electron current in the emitter

Q16. The region of trasistor in which extra impurity is doped to obtain a large number of majority carrier is called as :–

(A) emitter

(B) base

(C) collector

(D) any one of these depending upon the trasistor

Q17. In a transistor :–

(A) The emitter has the least concentration of impurity

(B) The collector has the least concentration of impurity

(C) The base has the least concentration of impurity

(D) All the three regions have equal concentration of impurity

Amplifier and switch

Q18. A common Emitter circuit is used as an amplifier, whose current gain is 50. If input resistance is 1kΩ and input voltage is 5 volt then output current will be :–

(A) 250 mA (B) 30 mA

(C) 50 mA (D) 100 mA

Q19. A transistor is used in the common emitter mode as an amplifier then :–

(A) The base emitter junction is forward baised.

(B) The base emitter junction is reverse baised.

(C) The input signal is connected in series with the voltage applied to bias the base emitter junction.

(D) The input signal is connected in series with the voltage applied to bias the base collector junction.

(A) A, B (B) A, D

(C) A, C (D) only C

Q20. In CB configuration of transistor ac current gain is $\frac{\Delta i_C}{\Delta i_E} = 0.98$ determine current gain of CE configuration :–

(A) 49 (B) 98

(C) 4.9 (D) 24.5

Q21. For a transistor amplifier power gain and voltage gain are 7.5 and 2.5 respectively. The value of the current gains will be :–

(A) 0.33 (B) 0.66

(C) 0.99 (D) 3

Logic gates

Q22. In Boolean algebra Y = A + B implies that

(A) Output Y exists when both input A and B exist

(B) Output Y exists when either input A exists or input B exists or both inputs A and B exist

(C) Output Y exists when either input A exists or input B exists but not when both inputs A and B exist.

(D) output Y exists when both inputs A and B exist but not when either input A or B exists.

Q23. The truth table given above is for which of the following gates :–

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

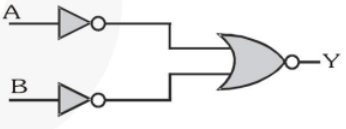
(A) NOR gate

(B) AND gate

(C) OR gate

(D) NAND gate

Q24. Which logic gate is represented by the following combination of logic gate :-



(A) OR

(B) NAND

(C) AND

(D) NOR

Q25. In the Boolean algebra, the following one which is not equal to A is-

(A) A.A

(B) A + A

(C) $\overline{A}.A$

(D) $\overline{\overline{A}+\overline{A}}$

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Directions for Assertion & Reason Questions

These questions Consist of two Statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(C) If Assertion is True but the Reason is false.

(D) If both Assertion & Reason are false.

Q1. **Assertion :** If the temperature of a semiconductor is increased, its resistance decreases.

Reason : The energy gap between conduction band and valence band in case of semiconductor is small.

(A) A (B) B

(C) C (D) D

Q2. If a battery of 5V is connected across a plate of the semiconductor of area 2×10^{-4} m^2 and of thickness 1.2×10^{-3} m, then the current flowing through the plate is :-

(A) 0.32 mA (B) 0.64 mA

(C) 1.28 mA (D) 2.56 mA

Directions for Assertion & Reason Questions

These questions Consist of two Statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(C) If Assertion is True but the Reason is false.

(D) If both Assertion & Reason are false.

Q3 **Assertion :** Germanium is preferred over silicon for making semiconductor devices.

Reason : Energy gap for Ge is more than that of Si.

(A) A (B) B

(C) C (D) D

Q4. If number of holes and free electrons in semiconductor are n_p and n_e respectively then :–

(A) $n_p > n_e$ in intrinsic semiconductor

(B) $n_p = n_e$ in extrinsic semiconductor

(C) $n_p = n_e$ in intrinsic semiconductor

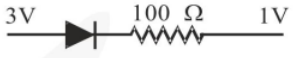
(D) $n_e > n_p$ in intrinsic semiconductor

Q5. For pure "Ge" semiconductor quantity of "e" and hole is 10^{19} e/m^3, if we doped donor impurity in it with density 10^{23} e/m^3 then quantity of hole (e/m^3) in semiconductor is:-

(A) 10^{15} (B) 10^{19}

(C) 10^{23} (D) 10^{27}

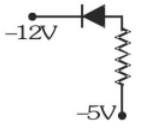
Q6. Assuming that the junction diode is ideal the current through the diode is :-

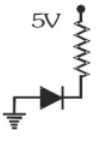


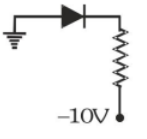
(A) 200 mA (B) 20 mA

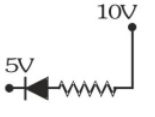
(C) 2 mA (D) zero

Q7. In given diagram which p-n junction is reverse biased

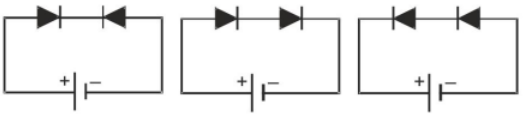
(A)


(B)


(C)


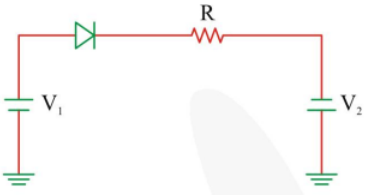
(D)


Q8. Two indentical P-N Jn. may be connected in series with a battery in three ways (fig below). the potential drops across the two P-N Jn. are equal in :–



(A) circuit 1 and 2 (B) circuit 2 and 3
(C) circuit 3 and 1 (D) circuit 1 only

Q9. If $V_1 > V_2$, r is resistance offered by diode in forward bias then current through the diode is :-



(A) 0 (B) $\frac{V_1 + V_2}{R + r}$

(C) $\frac{V_1 - V_2}{R + r}$ (D) None

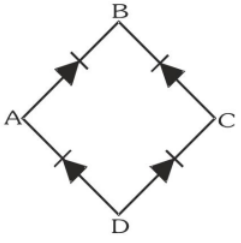
Q10. For a junction diode , the ratio of forward current (I_f) and reverse current is :

[I_e = electronic charge,
V = voltage applied across junction,
k = Boltzmann constant
T = temperature in kelvin]

(A) $e^{-V/kT}$ (B) $e^{V/kT}$

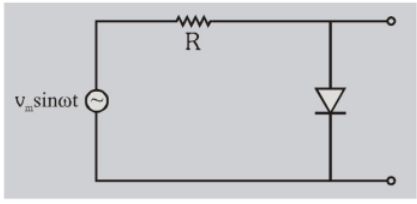
(C) $\left(e^{eV/kT} - 1\right)$ (D) $\left(e^{V/kT} - 1\right)$

Q11. In the figure, input is applied across A and C and output is taken across B and D, then the output is :



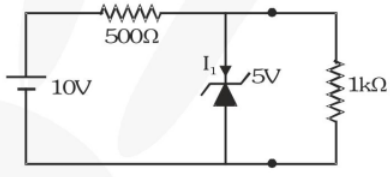
(A) zero
(B) same as input
(C) full wave rectified
(D) half wave rectified

Q12. The output across the diode in the given circuit in figure.



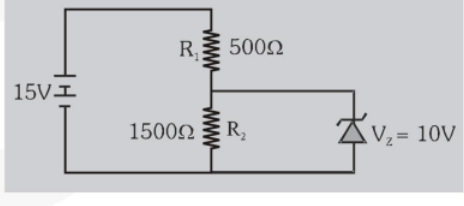
(A) Would be zero at all times.
(B) Would be like a half wave rectifier with positive cycles in output.
(C) Would be like a half wave rectifier with negative cycles in output.
(D) Would be like that of a full wave rectifier.

Q13. The current flowing through the zener diode in figure is :–



(A) 20 mA (B) 25 mA
(C) 15 mA (D) 5 mA

Q14. In the circuit given the current through the zener diode is :-



(A) 10 mA (B) 6.67 mA
(C) 5 mA (D) 3.33 mA

Q15. In an n-p-n transistor circuit, the collector current is 20 mA. If 90% of electron emitted reach the collector :–

(A) the emitter current will be 18 mA
(B) emitter current will be 22 mA
(C) base current will be 2 mA
(D) base current will be 1 mA

Q16. The current of transistor in common emitter mode is 49. The change in collector current and emitter current corresponding to the change in the base current by 5.0 μA, will be :–

(A) 245 μA, 250 μA (B) 240 μA, 235 μA
(C) 260 μA, 255 μA (D) none of these

Q17. For transistor current relation is :-

(A) $\alpha = \frac{\beta}{1-\beta}$ (B) $\beta = \frac{\alpha}{1-\alpha}$

(C) $\alpha = \frac{1+\beta}{\beta}$ (D) $\beta = \frac{1+\alpha}{\alpha}$

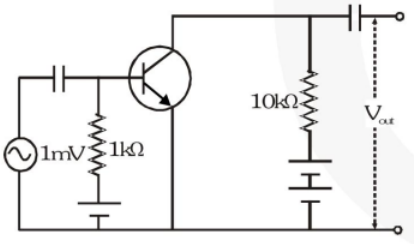
Q18. npn transistors are preferred to pnp transistors because they have

(A) low cost
(B) low dissipation energy
(C) capable of handling large power
(D) electrons have high mobility than holes and hence high mobility of energy.

Q19. In a common emitter amplifier, the phase difference between the input signal voltage and output voltage is :-

(A) $\frac{\pi}{2}$ (B) 0 (C) π (D) $\frac{\pi}{4}$

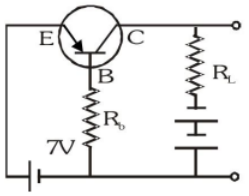
Q20. In the following common emitter configuration an 'npn' transistor with current gain β = 100 is used. The output voltage of amplifier will be :-



(A) 10 mV (B) 0.1 V
(C) 1.0 V (D) 10 V

Q21. In the given transistor circuit, the base current is 35 μA. The value of R_b is V_{BE} is assumed to negligible :-

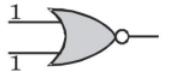
(A) 100 kΩ
(B) 300 kΩ
(C) 200 kΩ
(D) 400 kΩ

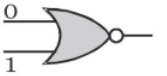


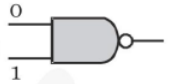
Q22. In a transistor amplifier β = 62, R_L = 5000 Ω and internal resistance of the transistor is 500 Ω. Its power amplification will be :

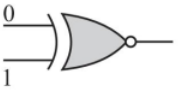
(A) 25580 (B) 33760
(C) 38440 (D) 55280

Q23. Which of the following gates will have an output of 1 :–

(a) 

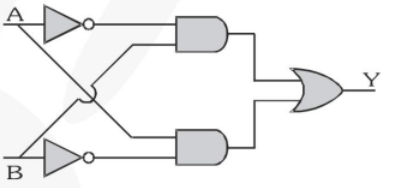
(b) 

(c) 

(d) 

(A) (a) and (b) (B) (b) and (c)
(C) (c) and (d) (D) (a) and (d)

Q24. Which of the following represents correctly the truth table of configuration of gates shown here



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

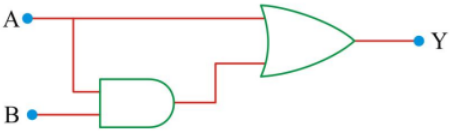
(C)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Q25. The output of the combination of the gates shown in the figure below is :-



(A) A + A. B (B) $A.(A+B)$
(C) $(A.B)+(\overline{A.B})$ (D) $(A+B)(\overline{A.B})$

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	A	A	B	D	D	A	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	A	B	D	A	A	C	A	C	A
21	**22**	**23**	**24**	**25**					
D	B	C	C	C					

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	D	C	A	B	B	B	B	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	D	D	D	A	B	D	C	C
21	**22**	**23**	**24**	**25**					
C	C	C	C	A					

-: Important Notes & Formulas :-